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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

MANAGEMENT OF TOTAL SPINAL ANAESTHESIA

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1. Introduction and Purpose

Total spinal anaesthesia is an iatrogenic anaesthetic emergency caused usually by the inadvertent injection of a large quantity of local anaesthetic into the subarachnoid space. It results from the action of local anaesthetic on both the spinal cord and brainstem.

2. Scope

This guideline is intended as a reference for trainee anaesthetists seeking advice regarding the management of total spinal anaesthesia.

3. Points to consider

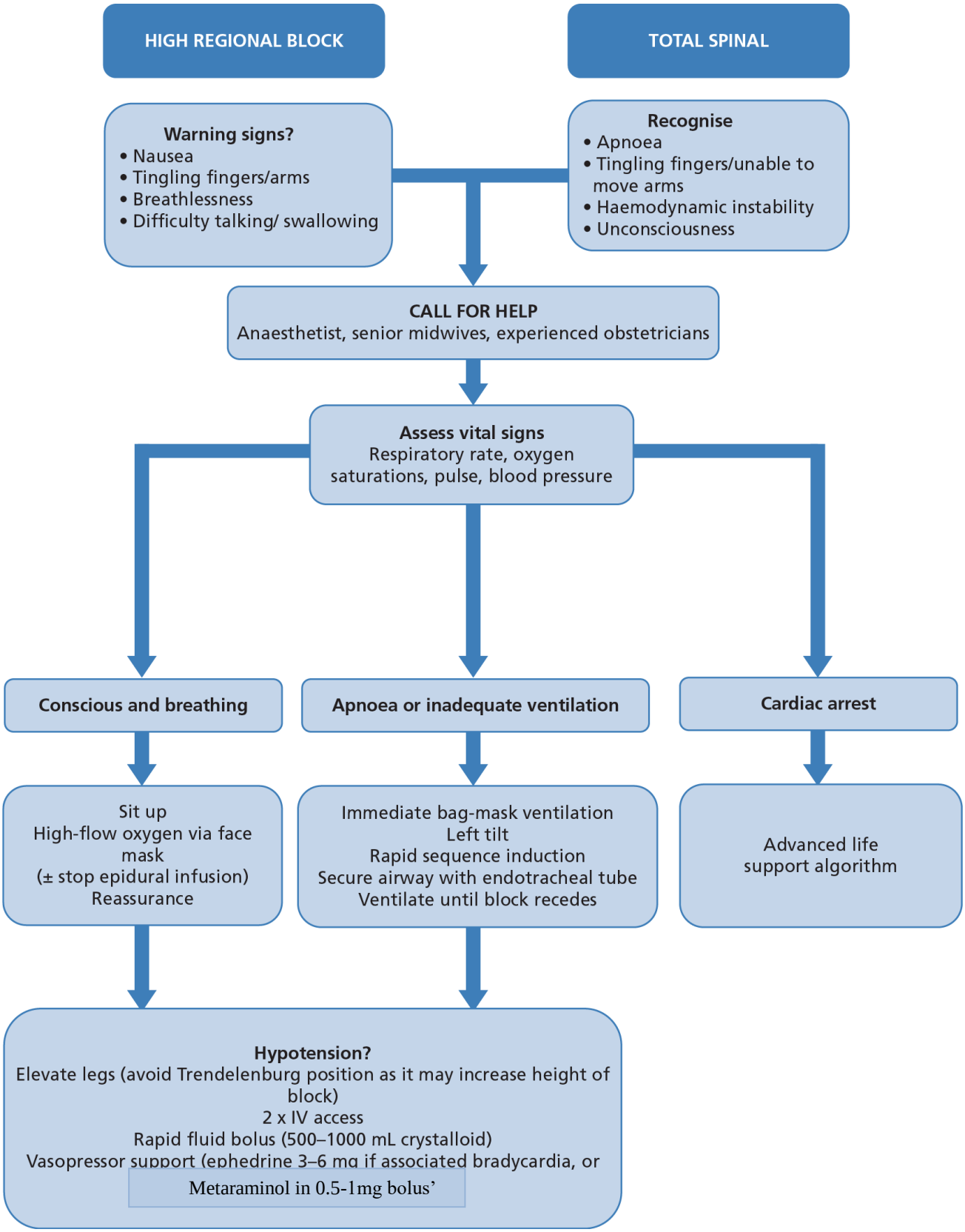
3.1 Recognition

- Usually occurs when an epidural top up e.g. for Caesarean Section is performed using an epidural catheter which has inadvertently been sited, or has migrated intrathecally

- A total spinal block can be differentiated from a high spinal block by worsening respiratory function

- Look for the following signs :
  a) Hypotension
  b) Bradycardia
  c) Numbness or weakness of the arms and hands
  d) Difficulty breathing
  e) Loss of consciousness

3.2 Management



- Ensure a 16G (or 14G) cannula is in situ with iv crystalloid or colloid running
- Maintain sedation with a propofol infusion (or volatile agent if the patient is already in theatre)
- Treat hypotension with a rapid infusion of 500 - 1,000ml of crystalloid or colloid, along with vasoconstrictors eg. metaraminol (0.5mg boluses) or ephedrine (6mg boluses)
- Treat bradycardia with atropine (600 microgram boluses)
- Consider inserting an arterial line for beat-to-beat blood pressure monitoring if the patient is cardiovascularly unstable, especially if inotropes are used
- Insert also a central line if inotropes are required
- Use a metaraminol infusion (or possibly a noradrenaline infusion) if the patient is persistently hypotensive. Use an adrenaline infusion if persistent hypotension is associated with persistent bradycardia despite treatment with atropine & ephedrine
- Monitor CTG
- Transfer the patient to the labour ward theatre if delivery / surgical intervention is necessary
- Following delivery / surgical intervention, transfer the patient to the intensive care unit for further supportive care and to await the spinal block to wear off.
- After a couple of hours the sedation can be discontinued. If the patient then wakes up and is able to breathe independently and adequately, then she can be extubated
- A full explanation of the events should be given to the patient, once awake

4. References

1. Epidural and Spinal anaesthesia. Chestnut D.H. (2004) *Obstetric Anaesthesia: Principles and Practice* Edition 3 Elsevier Mosby: 339
2. Algorithm for the Management of Total Spinal Block. *PROMPT Course Manual* Edition 3 (2017) Page 58